Pi4 Hat (PiHat) – Engineering and Service Document

Streamer DAC Raspberry Pi interface/HAT board

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1. Document information

Item	Value	Evidence
Product / board	Pi4 Hat (PiHat); KiCad project `Pi4Hat` inside `...\PIHat\Pi4Hat`	[K]
Function	Raspberry Pi 40-pin interface / HAT-style board for power distribution and signal break-out	[K]/[I]
KiCad title-block date	15 Nov 2012 on both schematic and PCB title blocks	[K]
KiCad source format	KiCad 8.0 files, analysed with `kicad-cli` 10.0.6; sources not modified	[K]
Source files used	`Pi4Hat.kicad_sch` and `Pi4Hat.kicad_pcb` (newest non-backup files in the supplied folder tree)	[K]
Existing summary reviewed	`boards\RaspberryPiInterfaceBoard.md`	[K]
Intended readers	Engineers, assemblers and service technicians	[I]

Evidence labels used throughout

Label	Meaning
[K]	Established from the KiCad schematic, PCB or KiCad-generated netlist/reports
[D]	Verified from a manufacturer / platform datasheet or official documentation read for this document
[C]	Calculated from [K]/[D] values
[I]	Inferred; plausible but not proven from the available files
[M]	Needs measurement or continuity testing on a physical board

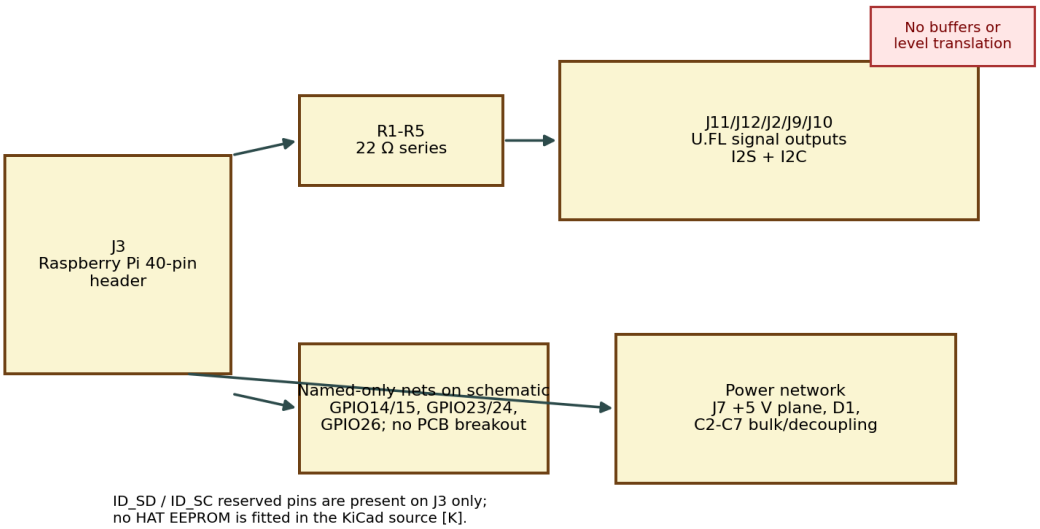
> **Scope and limits.** No physical board was measured and no firmware or software configuration was inspected. Nothing in this document is a hardware test result. Where a value is not in the project files it is stated as *not established from these files*. Open points are collected in OpenQuestions.md.

2. Overview

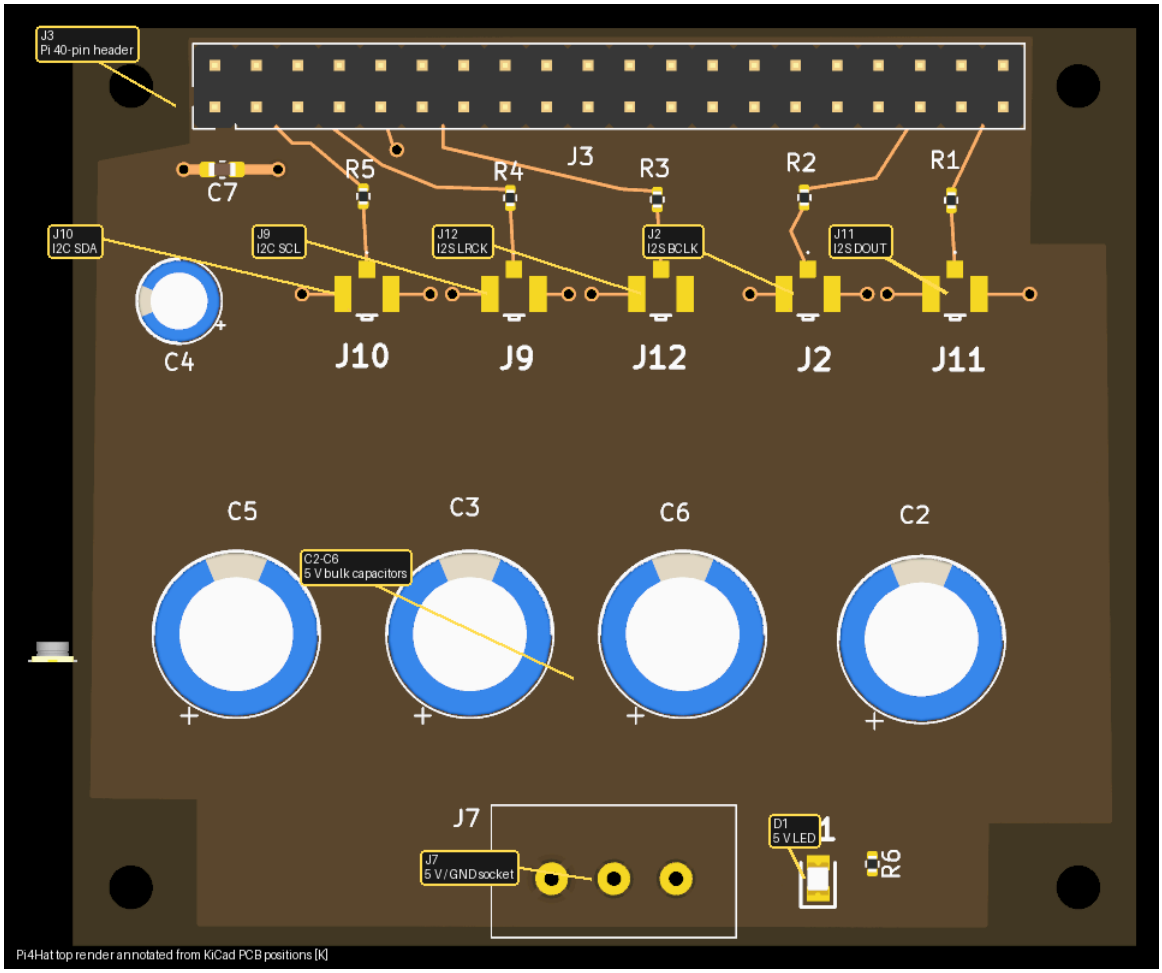
The Pi4 Hat is a passive Raspberry Pi 40-pin break-out board. In the KiCad source it does **not** contain logic ICs, regulators or level shifters [K]. Its main functions are:

- bring Raspberry Pi signals from the 40-pin header (J3) to five U.FL connectors through 22 Ω series resistors: I^2S data, I^2S frame-sync, I^2S bit-clock, I^2C SDA and I^2C SCL [K];
- provide local 5 V bulk capacitance and a 5 V present LED on one +5 V net [K];
- provide a 3-pin socket (J7) for that +5 V net [K];
- carry several named GPIO nets on the schematic for UART, handshake and PWM monitor lines, although these named nets are **not routed to separate PCB connectors in the current board file** [K];
- mechanically fit the Raspberry Pi 4 connector envelope; the PCB outline and silkscreen mark USB, RJ45, USB and PoE keep-out regions [K].

A major finding is that the KiCad source defines **two separate 5 V domains**: J7/C2-C7/D1/J4/J5 on net +5V, and Raspberry Pi header pins 2/4 with J8/J13 on net `Net-(J13-Pin_1)` [K]. The older summary document says J7 powers the Pi header, but the current KiCad source does not show that connection [K].



Functional block diagram



Annotated top view

3. Specifications

Parameter	Value	Evidence
PCB	4 layers, 1.6 mm FR-4 stack-up: F.Cu / In1.Cu / In2.Cu / B.Cu	[K]
Outline envelope	Approximately 65 mm × 56 mm, with non-rectangular Pi-clearance cut-outs	[K]
Mounting	4 × 2.7 mm holes for M2.5 hardware; positions match the usual Raspberry Pi HAT pattern	[K]/[I]
Pi connector	2 × 20, 2.54 mm header J3 (Molex 70246-4001 value in schematic)	[K]
External signal connectors	5 × U.FL connectors (J10 SDA, J9 SCL, J2 BCLK, J12 LRCK, J11 DOUT)	[K]
Series impedance on signal outputs	22 Ω in series with each exported signal (R1-R5)	[K]
Bulk capacitance on +5V	4 × 470 µF + 4.7 µF + 0.1 µF = 1884.8 µF total	[K]/[C]
Power indicator	D1 with R6 = 1 kΩ, connected to +5V	[K]
Local regulation	None found	[K]
Host logic levels	Raspberry Pi GPIO outputs are 3.3 V; inputs are 3.3 V-tolerant	[D]

Parameter	Value	Evidence
Fixed I²C pull-ups on Pi	GPIO2 and GPIO3 include fixed pull-ups to 3.3 V on the Pi	[D]
HAT ID EEPROM	Not fitted in these files	[K]
Operating current, temperature rise, cable loss	Not established from these files	–

Design checks

Check	Result	Evidence
Pi GPIO voltage compatibility	The U.FL signal outputs are direct Raspberry Pi GPIO nets through 22 Ω only. They are therefore 3.3 V logic outputs, **not 5 V tolerant interfaces**. External equipment must be high-impedance and 3.3 V compatible.	[K]/[D]
I²C assignment	J3 pin 3 = GPIO2/SDA1 and pin 5 = GPIO3/SCL1, matching Raspberry Pi I²C1 pins.	[K]/[D]
UART assignment	J3 pin 8 = GPIO14/TXD0 and pin 10 = GPIO15/RXD0, matching Raspberry Pi UART pins.	[K]/[D]
I²S assignment	J3 pin 12 = GPIO18 / PCM clock, pin 35 = GPIO19 / PCM frame-sync, pin 40 = GPIO21 / PCM data-out, consistent with Raspberry Pi PCM/I²S usage.	[K]/[D]
5 V routing intent	The KiCad source separates J7 +5V from Pi header pins 2/4. J7 therefore cannot be assumed to power the Pi in the current source set.	[K]
3.3 V usage	J3 pins 1 and 17 are both net `+3.3V`, but no local 3.3 V load is connected in the schematic.	[K]
Protection	No ESD, over-voltage or buffering is fitted between J3 and the exported signal connectors.	[K]

4. Connectors, controls, indicators and test points

4.1 J3 – Raspberry Pi 40-pin header [K]

J3 is the main board-to-host connection. Only a subset of pins is actively used by the current design.

J3 pin(s)	Net / function	Notes
1, 17	`+3.3V`	Present from the Pi only; no local 3.3 V load found [K]
2, 4	`Net-(J13-Pin_1)`	Pi 5 V pins; **not on the +5V net used by J7/C2-C7/D1** [K]
3	`GPIO2/SDA1`	Routed through R5 to J10 [K][D]
5	`GPIO3/SCL1`	Routed through R4 to J9 [K][D]
6, 9, 14, 20, 25, 30, 34, 39	GND	Main ground pins used in the design [K]

J3 pin(s)	Net / function	Notes
8	GPIO14/TXD0	Named on the schematic only; no separate PCB output found [K]
10	GPIO15/RXD0	Named on the schematic only; no separate PCB output found [K]
12	GPIO18/I2S BCL	Routed through R3 to J2 [K][D]
16	GPIO23 /ESP DRY	Named on the schematic only [K]
18	GPIO24/RPI DRY	Named on the schematic only [K]
35	GPIO19/I2S LRCK	Routed through R2 to J12 [K][D]
37	GPIO26/Mon PWM	Named on the schematic only [K]
40	GPIO21/I2S DOUT	Routed through R1 to J11 [K][D]
27, 28	ID_SDA, ID_SCL	Reserved HAT EEPROM pins; schematic note says do not use except for an I^2C ID EEPROM [K]

All remaining J3 pins are no-connect or have only symbolic labels in the schematic [K].

4.2 U.FL signal connectors [K]

Each U.FL connector exports one signal plus two ground pads. No active buffering is provided.

Ref	Signal	Source pin
J10	I^2C SDA	J3 pin 3 / GPIO2 via R5 22 Ω
J9	I^2C SCL	J3 pin 5 / GPIO3 via R4 22 Ω
J2	I^2S BCLK / PCM clock	J3 pin 12 / GPIO18 via R3 22 Ω
J12	I^2S LRCK / PCM frame-sync	J3 pin 35 / GPIO19 via R2 22 Ω
J11	I^2S DOUT / PCM data-out	J3 pin 40 / GPIO21 via R1 22 Ω

These connectors should be treated as 3.3 V logic-level outputs only [K][D]. Characteristic impedance, cable length limits and downstream terminations are *not established from these files*.

4.3 J7 – 5 V / GND socket [K]

Pin	Net
1	+5V
2	+5V
3	GND

The current KiCad source ties J7 to the local +5V plane with C2-C7 and D1, but **not** to J3 pins 2/4 [K]. Whether J7 is intended as an input, output or link point must be confirmed on hardware [M].

4.4 Controls, indicators and test points [K]

Item	Function
D1	5 V present LED on local +5V net
R6	1 kΩ current-limit resistor for D1
Test points / switches	None found

4.5 Schematic-only connectors not present on the current PCB [K]

The schematic includes the following connectors, but the PCB file does not contain footprints for them:

Ref	Schematic value	Net / purpose
J1	`GPIO`	Unconnected 2 × 20 header placeholder
J4, J5	`5v pin`	Single-pin connectors on local +5V
J6	`1725672`	4-pin ground terminal
J8, J13	`5v pin`	Single-pin connectors on `Net-(J13-Pin_1)`
J14	`5v pin`	Single-pin connector on isolated `GND2`

This schematic/PCB mismatch must be resolved before the design can be treated as production-ready [K][I].

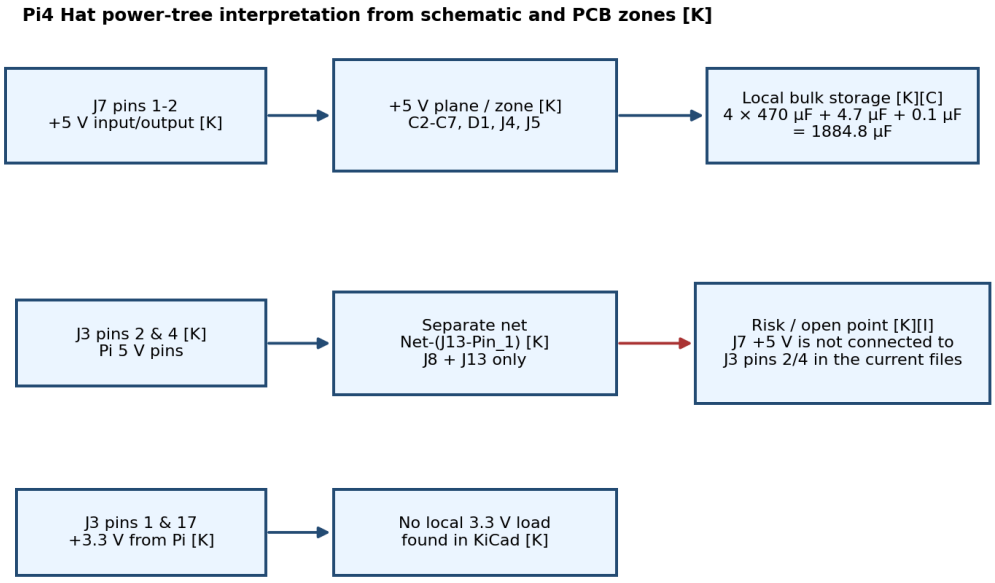
5. Functional description

5.1 Signal break-out

The board directly exports Raspberry Pi signals to coax connectors. R1-R5 (all 22 Ω) are the only components between the host header and the U.FL outputs [K]. Therefore:

- no level translation is present [K];
- no active buffering is present [K];
- no dedicated ESD protection is present [K];
- any downstream board must be 3.3 V compatible and should not force these nets above 3.3 V [D].

5.2 Power distribution



Power tree

Two separate power networks are described in the KiCad source [K]:

1. **Local +5V plane** on In2.Cu feeding J7, C2-C7, D1 and schematic-only J4/J5.

2. **`Net-(J13-Pin_1)`** feeding J3 pins 2 and 4 and schematic-only J8/J13.

The old summary document states that J7 feeds the Raspberry Pi 5 V pins, but that is **not** what the present schematic/netlist says [K]. This is the single most important service ambiguity on the board.

5.3 Reserved and named-only nets

The schematic note explicitly warns not to use `ID_SD` and `ID_SC` for anything other than an I^2C ID EEPROM [K]. No EEPROM is fitted in the current KiCad source [K], so automatic HAT identification is not implemented [I].

Several additional GPIO nets are named on J3 (`GPIO14/TXD0`, `GPIO15/RXD0`, `GPIO23/ESP DRY`, `GPIO24/RPI DRY`, `GPIO26/Mon PWM`) but have only one connection in the KiCad netlist and no dedicated PCB connector [K]. They should therefore be treated as documentation labels, not usable external interfaces, unless a later revision adds the missing footprints/routes [I].

5.4 Mechanical intent

The board outline is shaped around the Raspberry Pi connector envelope, and the silkscreen marks `USB`, `RJ45`, `USB` and `PoE` keep-out references [K]. This strongly suggests the board is intended to sit above a Raspberry Pi 4 [I]. The DRC nevertheless reports a malformed outline, so the mechanical definition should be checked before manufacture [K].

6. Assembly and commissioning

6.1 Before assembly / first fit

1. Confirm whether the current product to be built should follow the **schematic**, the **PCB**, or a later unpublished revision. Seven schematic connectors are absent from the PCB file [K].
2. Decide which 5 V net is intended to power the Raspberry Pi. In the current source J7 does **not** connect to J3 pins 2/4 [K].
3. Check that the chosen U.FL cables and downstream circuitry are compatible with 3.3 V GPIO signalling [K][D].
4. Verify electrolytic capacitor polarity (C2, C3, C5, C6) and LED polarity before soldering [K].
5. Check J3 orientation carefully; the board is keyed only by the 2 × 20 pin pattern and mounting holes [K].

6.2 Suggested continuity checks before power [M]

These are recommended checks, **not measured results**:

Check	Expected from KiCad
J7 pin 1 to D1 anode	Continuity on local `+5V`
J7 pin 1 to J3 pin 2	**Open circuit** if the present net split is real
J3 pin 3 to J10 SIG	22 Ω through R5
J3 pin 5 to J9 SIG	22 Ω through R4
J3 pin 12 to J2 SIG	22 Ω through R3
J3 pin 35 to J12 SIG	22 Ω through R2
J3 pin 40 to J11 SIG	22 Ω through R1
J14 to main GND	Open circuit in the current schematic (`GND2` is isolated)

6.3 First power-up guidance [M]

1. If powering the board standalone through J7, verify 5 V at D1 and across the bulk capacitors.
2. **Do not assume this also powers the Raspberry Pi** until continuity from J7 to J3 pins 2/4 is confirmed.
3. When fitting to a Raspberry Pi, confirm the host still boots normally and that the exported GPIO nets idle at 3.3 V-compatible levels.
4. Enable and configure the Pi interfaces in software as needed (I^2C, PCM/I^2S, UART). No automatic HAT EEPROM configuration is present in the KiCad files.

7. Testing and troubleshooting

Symptom	Likely cause	Check
Pi does not power from J7	KiCad source splits J7 +5V from J3 pins 2/4	Continuity J7 → J3 pin 2/4 [K][M]
D1 off but Pi still powered	Pi 5 V may be arriving on `Net-(J13-Pin_1)` rather than local +5V	Check which 5 V net is energised [K][M]
No I^2C on J10/J9	Pi I^2C disabled, wrong cable, or damaged R4/R5 / U.FL joints	Check GPIO2/GPIO3 activity and 22 Ω continuity
No audio data on J11/J12/J2	Pi PCM/I^2S disabled or wrong device-tree configuration	Check GPIO18/19/21 at J3 first, then across R1-R3
UART / handshake lines unavailable	These nets are named in the schematic only and not broken out on the PCB	Review §4.5 before fault-finding
Noise or damaged GPIO	External device is not 3.3 V compatible; no buffer or ESD protection fitted	Disconnect downstream cable and inspect Pi GPIO function
J14 has no usable voltage	`GND2` is isolated in the current schematic	Continuity J14 → GND should be open [K][M]
One U.FL ground shell not bonded as expected	DRC reports one unconnected ground item at J12	Inspect J12 ground pads and copper around them [K][M]

ERC summary: 41 errors and 12 warnings [K]. DRC summary: 23 rule violations plus 5 unconnected-pad items [K]. Many of the ERC errors are caused by the unconnected placeholder header J1 and by named nets with no second connection; they still indicate that the design files need cleanup before release [K][I].

8. Repair and rework

- U.FL connectors are delicate; use suitable hot-air or fine-tip rework tools and inspect pad lift afterwards [I].
- Replace R1-R5 only with the intended 22 Ω value unless signal-integrity testing justifies a change [K].
- Replace R6 only with a value that keeps LED current modest at 5 V; 1 kΩ is fitted in the KiCad source [K].
- Observe electrolytic polarity when replacing C2/C3/C5/C6 [K].
- If correcting the design rather than repairing a board, resolve the schematic/PCB mismatch before making any service judgement from the files [K].

9. BOM reference

9.1 PCB-fitted items in the current board file [K]

Ref	Qty	Value	Footprint / note
C2, C3, C5, C6	4	470 µF	THT radial electrolytic

Ref	Qty	Value	Footprint / note
C4	1	4.7 µF	THT radial capacitor
C7	1	100 nF	0603
D1	1	LS_R976	LED, 0805-style footprint
R1-R5	5	22 Ω	0402
R6	1	1 kΩ	0402
J3	1	70246-4001	2 × 20 Raspberry Pi header
J7	1	Conn_01x03_Socket	3-pin 5 V / GND socket
J2, J9, J10, J11, J12	5	U.FL-R-SMT-1_80_	U.FL signal connectors
MH1-MH4	4	MountingHole_2.7mm_M2.5	PCB-only mechanical features

9.2 Schematic-only items missing from the PCB file [K]

Ref	Value	Comment
J1	GPIO	Placeholder 2 × 20 connector, fully unconnected
J4, J5	5v pin	Single-pin +5V connectors
J6	1725672	4-pin ground terminal
J8, J13	5v pin	Single-pin connectors on Pi 5 V net
J14	5v pin	Single-pin connector on isolated `GND2`

10. Limitations, risks and open points

1. **5 V routing ambiguity.** The current KiCad source does not connect J7 +5V to Raspberry Pi header pins 2/4 [K]. This directly contradicts the older summary and must be resolved.
2. **Schematic/PCB mismatch.** Seven schematic connectors are absent from the PCB, and four mounting holes exist only on the PCB [K].
3. **No buffering or protection.** Exported signals are direct Pi GPIO nets through 22 Ω only [K][D].
4. **J14 is misleading.** It is labelled `5v pin` in the schematic but is actually on isolated `GND2` [K].
5. **No HAT EEPROM.** The reserved ID pins are present, but no ID EEPROM is fitted [K].
6. **Design-rule state is not clean.** ERC and DRC both report substantive issues, including malformed outline, unconnected items and clearance/thermal problems [K].
7. **Old summary discrepancies.** The older `RaspberryPiInterfaceBoard.md` summary states that J7 powers the Pi header and does not mention the schematic/PCB part mismatch [K].

See OpenQuestions.md for the short action list.

11. References and revision history

11.1 References

- KiCad source: `C:\Users\antho\Documents\StreamerDac Project\Documents\kicad\Streamer schematic\ControlBoardTiny\tinyControlBoard\PIHat\Pi4Hat\Pi4Hat.kicad_sch` and `.kicad_pcb` [K]
- KiCad-generated netlist, ERC and DRC reports created for this document [K]
- Existing summary: `docs\userDocumentation\service-guide\boards\RaspberryPiInterfaceBoard.md` [K]

- Raspberry Pi official GPIO documentation (`gpio-on-raspberry-pi.adoc`) for voltage levels and fixed pull-ups [D]
- pinout.xyz pages for physical-pin to GPIO/alternate-function cross-checks on GPIO2, GPIO3, GPIO14, GPIO15, GPIO18, GPIO19 and GPIO21 [D]

11.2 Revision history

Revision	Date	Change
A	2026-10-11	First engineering and service document generated from the supplied KiCad source